



TAEKWONDO KICKMETER

Project Artefact Report



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1. Introduction

The Taekwondo Kickmeter is an embedded IoT solution designed to objectively measure the force, speed, and accuracy of Taekwondo kicks using integrated sensors and wireless communication. Traditional martial arts training relies heavily on visual observation and subjective feedback, often lacking quantitative metrics. This project addresses that gap by creating a data-driven, real-time feedback system for athletes and coaches.

The system integrates multiple sensing technologies — a load cell for measuring impact force, FSR sensors for contact accuracy, and an IMU (Inertial Measurement Unit) from an Arduino Nano 33 IoT for motion dynamics. These sensors communicate via BLE (Bluetooth Low Energy) and are processed on a Raspberry Pi, which uploads the data to a Firebase Realtime Database. The results are displayed through a React-based web dashboard, allowing immediate visualization of each kick's performance.

The Kickmeter demonstrates principles of embedded systems design, IoT communication, and real-time data processing. The prototype emphasizes robustness, fault tolerance, and responsiveness, meeting the High Distinction criteria outlined in the project rubric. Beyond a functioning prototype, the project delivers a well-documented, replicable solution capable of extending to other combat sports or physical performance measurement systems.

2. Project Overview

1.1 Background

Taekwondo, originating in Korea in the mid-20th century, is a martial art that emphasizes speed, power, precision, and discipline. It combines elements of traditional Korean combat styles with modern athletic training, and has evolved into a globally practiced Olympic sport. The effectiveness of a Taekwondo kick depends not only on technique but also on measurable factors such as impact force, reaction time, and accuracy of contact.

Traditionally, performance evaluation in Taekwondo relies on subjective observation and visual feedback from coaches, which makes it challenging to track consistent, measurable progress. However, with rapid advancements in embedded systems, sensor technologies, and Internet of Things (IoT) solutions, it is now possible to collect and analyze training data in real time.

This project — the Taekwondo Training Sensor System (Kickmeter) — was inspired by a personal motivation to blend Taekwondo and technology. It aims to quantify performance metrics such as kick strength, speed, and accuracy, providing objective, data-driven feedback for athletes. By leveraging IoT principles and multi-sensor integration, this system helps transform traditional martial arts training into a scientifically measurable, modern performance tool.

1.2 Existing Work

Research and commercial efforts in the domain of martial-arts and kick-impact systems show that while there is relevant work, several important gaps remain.

- There are commercial solutions, for example the Daedo “PSS Scoring System” used in elite taekwondo competitions, which uses high-sensitivity force sensors integrated into protective gear to detect valid strikes (force thresholds, wireless transmitters, real-time scoring). However, these systems are primarily designed for scoring in a competition context (detecting valid hits) rather than providing multi-metric training feedback (speed, accuracy, force) or being easily customizable by end-users (coaches/athletes) for research or training.
- In the DIY / academic realm, many projects focus on measuring either force or speed, but not a full multi-sensor fusion of speed + force + accuracy. For example, one system described in 2007 measured both speed and force of kicks using a target board with photoelectric switches and pressure sensors, but it was a laboratory setup and did not integrate a full real-time, cloud-based system. Another study developed a wearable IMU sensor module for kick detection and impact level monitoring, but focused on identifying anomalous kick types rather than an integrated training feedback system fusing multiple sensors and visualisation.
- There are few systems that combine real-time cloud data upload, multi-sensor fusion (force + speed + accuracy), and visualisation/dashboarding for athlete feedback. For instance, a review of sports biomechanics sensors notes that while force plates and accelerometers exist, the translation into scalable, real-world athlete training systems remains limited.

Thus, while meaningful foundation exists, the landscape shows that most training-oriented systems are either expensive, limited in scope (e.g., force only or speed only), or lack full real-time feedback + customisability. This leaves a clear opportunity for a project like yours to fill the gap by delivering a cost-effective, multi-sensor, real-time feedback system for taekwondo training.

1.3 Problem Statement

Taekwondo athletes require precise feedback on their performance to improve effectively. However, existing solutions are often expensive, bulky, and limited in functionality, measuring only force, speed, or pressure individually. Most systems fail to provide real-time, quantitative feedback, making it difficult to track progress accurately during training.

There is a clear need for a system that is:

- **Affordable** – accessible to athletes and training centers.
- **Portable** – easy to deploy in various training environments.
- **Accurate and Comprehensive** – capable of simultaneously measuring kick force, speed, and pressure.
- **Real-Time** – providing instant, actionable feedback.
- **Customizable** – adaptable to individual training requirements.

The objective is to develop a reliable embedded system that addresses these gaps and enables effective, data-driven Taekwondo training.

1.4 Requirements

Hardware Requirements:

- **Raspberry Pi 4** – central processing and data handling.
- **Arduino Nano 33 IoT** – sensor interfacing and BLE communication.
- **HX711 Load Cell with KG measurement module** – for measuring kick force.
- **Wooden plates** – to mount the load cell securely.
- **FSR Sensors** – to capture foot pressure distribution.
- **ADS1115 ADC Module** – for high-resolution analog-to-digital conversion of sensors.
- **Power Bank** – to provide portable power to the system.
- **Jumper wires and connectors** – for flexible and reliable electrical connections.

Software Requirements:

- **Python** – for sensor integration, data acquisition, and processing.
- **React.js** – for building the frontend dashboard for real-time visualization.
- **Firebase** – for cloud storage and real-time data synchronization.
- **Fusion 3D** – for CAD design of the load cell structure.
- **RD Works** – for laser cutting the structural components designed in CAD.

Functional Requirements:

- Accurately measure kick force in kilograms and Newtons.
- Capture speed of kicks.
- Collect FSR pressure data to analyze foot placement and impact.
- Integrate data from all sensors and upload it to Firebase.
- Display real-time results on a dashboard for training analysis.

Non-functional Requirements:

- Ensure real-time data transmission with minimal latency.
- Maintain fault-tolerant BLE communication to prevent data loss.
- The system must be portable, robust, and easy to calibrate.

2. Design Principles

- **Modularity:** The system is designed with independent sensor modules (load cell, FSRs, BLE Arduino) that communicate via a central aggregator. This ensures maintainability, scalability, and testability, allowing individual components to be upgraded or replaced without affecting the overall system.
- **Asynchronous Communication:** Sensor readings and BLE data transmission are handled asynchronously, improving average and worst-case performance, reducing latency, and enhancing energy efficiency by avoiding blocking operations.

- **User-Centric Design:** Calibration procedures are simple and intuitive, and real-time visual feedback is provided through the React.js dashboard. This enhances usability and safety, ensuring users can easily understand and trust the system.
 - **Fault Tolerance:** The system incorporates automatic BLE reconnection, load cell signal averaging, and error detection mechanisms. These features improve reliability, numeric precision, and robustness against transient faults or communication errors.
 - **Scalability:** The architecture allows easy addition of new sensors or machine learning modules without major redesign. This supports long-term maintainability, performance optimization, and adaptability to evolving requirements.
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3. Prototype Architecture

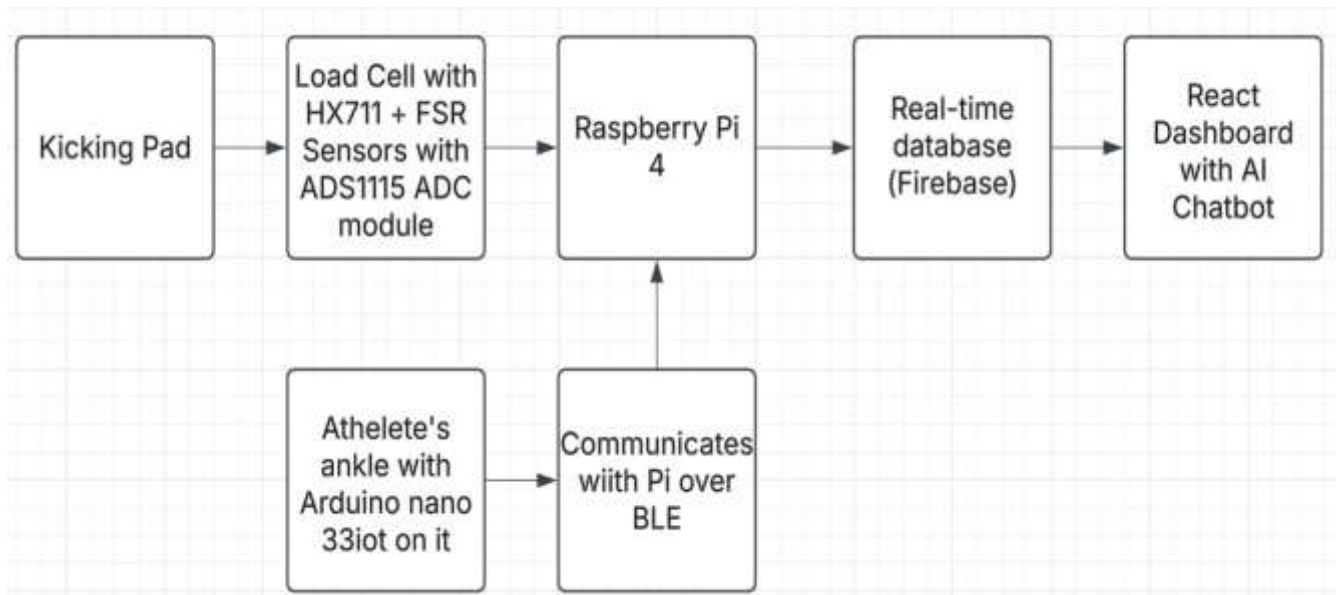
3.1 Hardware Architecture

The hardware architecture of the Kickmeter system integrates multiple sensors and processing units to capture comprehensive metrics of Taekwondo kicks:

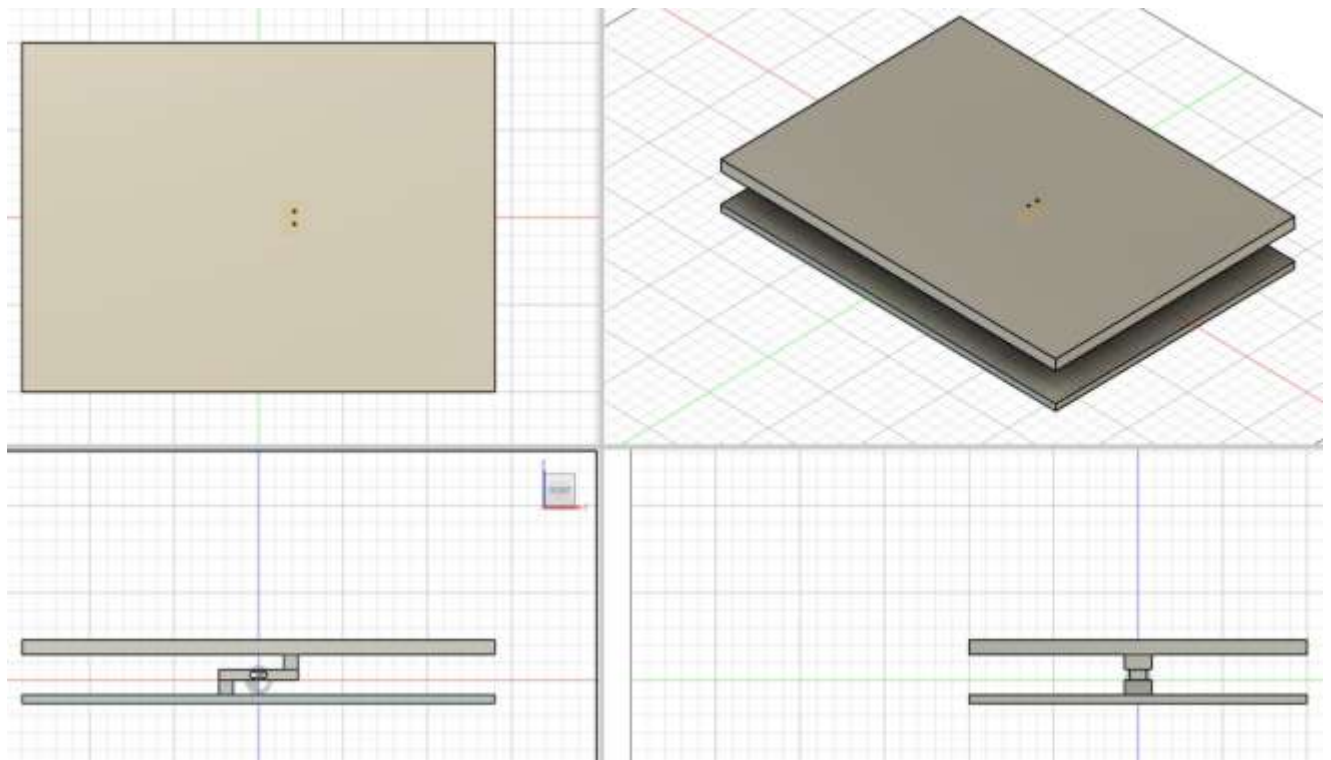
- **HX711 Load Cell:** The load cell is mounted between two laser-cut wooden plates using spacers on alternating edges. This arrangement ensures that forces applied on one side do not cancel out forces on the other, maintaining accurate measurement. The assembly is embedded within the kicking pad, with the load cell positioned 5–6 cm beneath the striking surface. It measures the impact force of each kick and is calibrated using known weights.
- **FSR Sensors:** Force-sensitive resistors are embedded beneath the surface of the kicking pad in a shallow arrangement (1–1.5 cm below the top surface). They detect pressure distribution across the pad and provide additional feedback on foot placement and strike quality. An ADS1115 ADC module converts analog signals from the FSRs into digital data for the Raspberry Pi.
- **Arduino Nano 33 IoT:** Worn on the ankle, the Arduino Nano 33 IoT captures foot acceleration and speed using its onboard accelerometer. It communicates the motion data wirelessly via BLE to the Raspberry Pi.
- **Raspberry Pi 4:** Mounted at the back of the kicking pad, the Raspberry Pi aggregates data from the load cell, FSRs, and Arduino Nano 33 IoT. It performs preprocessing tasks such as filtering and kick detection, and uploads the combined data to Firebase in real time.
- **Power and Communication:** The Raspberry Pi is powered through its adapter and accessed via SSH from a laptop for development and monitoring. The Arduino Nano 33 IoT is powered by a separate power bank, enabling wireless operation during kicks.

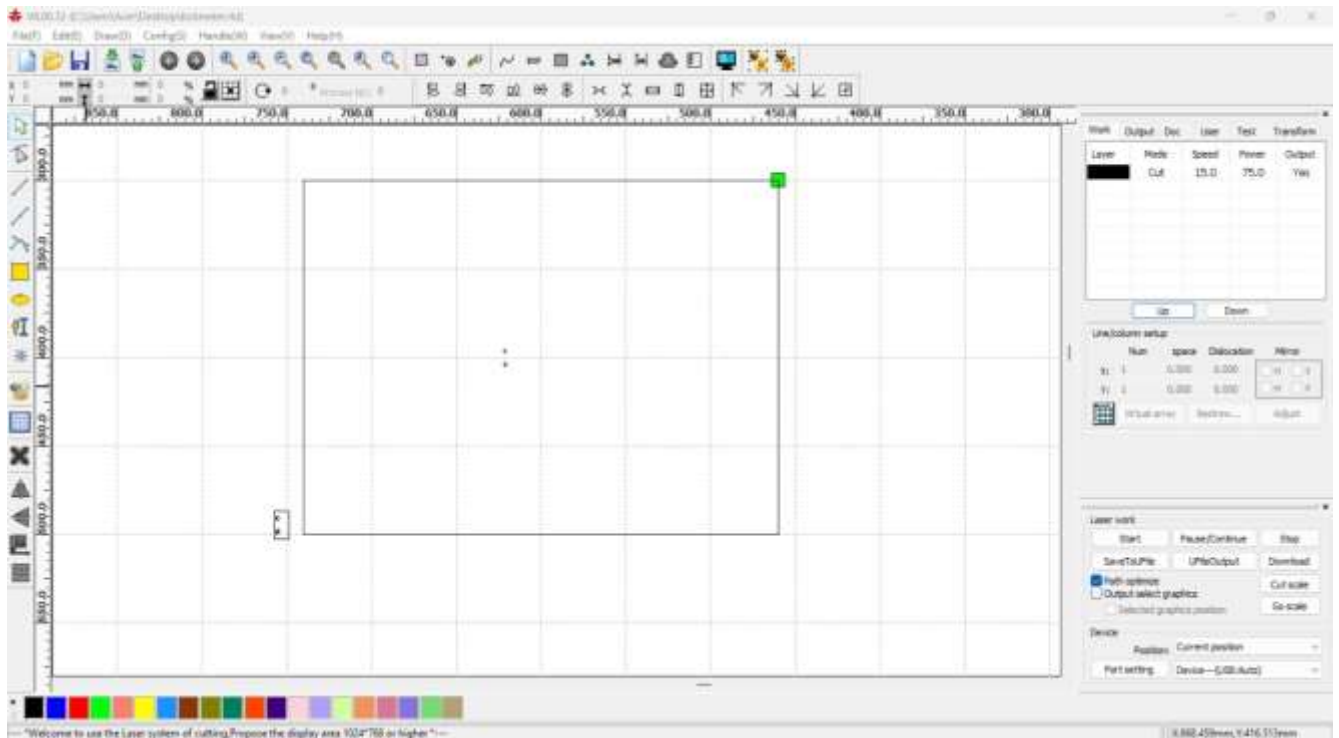
This modular architecture ensures reliable data acquisition, fault tolerance, and scalability, allowing the system to provide detailed performance insights after each training session. The data is visualized on a website via React.js, and a chatbot provides feedback and suggestions for improvement.

Architecture Diagram:



3.2 CAD Design of Load Cell Mount and RLD file:





3.3 Software Architecture

- **Sensor Integration Layer:** Python scripts on the Raspberry Pi read data from the HX711 load cell and FSR sensors. The layer handles calibration, signal averaging, and analog-to-digital conversion (via the ADS1115 ADC for FSRs), ensuring accurate and reliable measurements.
- **BLE Communication Layer:** Acceleration and speed data from the Arduino Nano 33 IoT is transmitted to the Raspberry Pi over Bluetooth Low Energy (BLE) using the Bleak Python library. The system incorporates automatic reconnection to handle temporary communication failures, improving fault tolerance.
- **Data Aggregation Layer:** Sensor readings from the load cell, FSRs, and Arduino accelerometer are combined into structured JSON objects. This layer performs preprocessing such as filtering, kick detection, and timestamp synchronization, preparing the data for cloud storage.
- **Cloud Integration Layer:** The Raspberry Pi uploads aggregated data to Firebase Realtime Database. This ensures persistent storage, enabling historical analysis of training sessions and supporting real-time updates for the frontend.
- **Frontend Layer:** A React.js dashboard visualizes live metrics such as kick force, pressure distribution, and speed. After a session of kicks, a chatbot provides automated insights and personalized suggestions. All session data and feedback are stored on the web platform for easy retrieval and analysis.

4. Prototype Code

- Full source code hosted on GitHub: <https://github.com/Simu000/Taekwondo-Kickmeter/tree/main>

- **Includes:**
 - **pi.py** – Raspberry Pi script that reads the HX711 load cell, FSR sensors, and fetches real-time acceleration data from the Arduino Nano 33 IoT over BLE. It combines all sensor data into structured JSON and uploads it to Firebase.
 - **accelerometer.ino** – Arduino Nano 33 IoT code that measures foot acceleration and speed using its onboard accelerometer and sends the data to the Raspberry Pi over BLE.
 - Sensor calibration scripts for both the load cell and FSRs.
 - BLE communication logic, including automatic reconnection for fault tolerance.
 - Data aggregation and preprocessing routines before cloud upload.
 - **Frontend files** – located under the src folder, containing the React.js dashboard and chatbot interface for real-time visualization and session feedback.
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5. Testing Approach

- **Unit Testing:** Each sensor module was verified individually. The HX711 load cell, FSR sensors, and Arduino Nano 33 IoT accelerometer were tested separately to confirm correct readings and proper communication with their respective interfaces.
 - **Integration Testing:** Combined system functionality was validated by ensuring seamless data flow from all sensors through the Raspberry Pi to Firebase. This verified that data aggregation, preprocessing, and cloud upload worked correctly in real-time.
 - **Calibration Testing:** The load cell was calibrated using known weights, resulting in an offset of 266305 and a scale factor of -180295.609375. FSR sensors were also tested to ensure accurate pressure mapping beneath the kicking pad surface.
 - **Stress Testing:** Extended kicking sessions were conducted to assess system stability and sensor performance over repeated use. BLE communication was monitored for latency and reliability under continuous operation.
 - **Fault Tolerance Testing:** BLE disconnections were simulated to verify that the system automatically reconnected and continued to collect and upload data without data loss. Load cell averaging and sensor error handling were also evaluated for robustness.
 - **User Testing:** Real Taekwondo training sessions were conducted to observe system feedback, verify accuracy of kick force, speed, and pressure distribution metrics, and ensure that the interface and chatbot provided meaningful insights for the user.
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6. User Manual

1. Hardware Setup

- The system is fully assembled:
 - HX711 load cell and FSR sensors embedded in the kicking pad.
 - Arduino Nano 33 IoT worn on the ankle.
 - Raspberry Pi 4 mounted at the back of the pad.

- Ensure both devices are powered:
 - Raspberry Pi via adapter
 - Arduino Nano 33 IoT via power bank

2. Software Operation

- The Raspberry Pi script pi.py handles all sensor readings, fetches Arduino data over BLE, and sends combined data to Firebase automatically.
- No additional software installation is required.

3. Calibration

- Calibration is already completed for both the load cell and FSRs.
- To recalibrate:
 - Delete the calibration.json file.
 - Run pi.py again; the system will perform calibration automatically.

4. Data Monitoring

- Open the website dashboard in a browser.
- Click “**Start Session**”.
- Wait a few seconds for the Arduino Nano 33 IoT to automatically connect to the Pi.
- Begin striking the pad. Live metrics will display:
 - Kick/strike force (load cell)
 - Pressure distribution (FSRs)
 - Foot acceleration and speed (Arduino Nano 33 IoT)
- After the session, the **chatbot** provides insights and suggestions. All session data and feedback are saved for future reference.

5. Troubleshooting & Safety Tips

- **BLE Reconnection:** Arduino reconnects automatically if disconnected; system resumes data collection.
- **Kick Force:** Avoid striking too hard. The load cell is rated for 10 kg; excessive force may cause damage.
- **Arduino Power:** Keep it on during sessions; turning it off stops data collection.
- **Raspberry Pi Shutdown:** Always shut down the Pi properly to prevent SD card corruption.

6. Quick Start

1. Power on the Raspberry Pi and Arduino Nano 33 IoT.
2. Open the website dashboard and click “**Start Session**”.
3. Wait for Arduino-Pi connection.
4. Begin training; view real-time metrics.
5. Review insights from the chatbot after the session.

This version ensures safe, easy operation while leveraging automatic calibration, data aggregation, and real-time feedback features.

7. Project Demonstration Video

- **Link:** <https://youtu.be/upUZykV8Sd8>
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8. Challenges and Failures

During the development of the Kickmeter system, several technical and integration challenges were encountered, which required iterative problem-solving and careful design adjustments:

- **Sensor Calibration:**
 - Initially, the HX711 load cell readings were inconsistent due to improper mounting and signal noise. Correctly mounting the load cell between laser-cut wooden plates with spacers on alternating edges stabilized the readings and prevented force cancellation.
 - FSR sensors exhibited non-linear responses, requiring careful placement beneath the kicking pad surface (1–1.5 cm) and calibration using known pressures to obtain reliable measurements.
- **BLE Communication:**
 - The Arduino Nano 33 IoT occasionally disconnected during sessions, interrupting real-time data flow. Automatic reconnection logic was implemented in the Raspberry Pi script to ensure the system resumed operation without manual intervention.
- **Raspberry Pi Stability:**
 - SSH sessions with the Raspberry Pi dropped intermittently during prolonged testing, affecting monitoring and debugging. This was mitigated by ensuring stable power supply, avoiding unnecessary heavy computation on the Pi, and using proper shutdown procedures to prevent SD card corruption.
- **Software Integration:**
 - Combining three sensor streams—load cell, FSRs, and Arduino accelerometer—initially caused delays and data mismatches. The Python code was refactored to use asynchronous programming, enabling simultaneous reading of sensors and BLE communication without blocking.
 - Preprocessing, such as averaging load cell readings and filtering FSR signals, was necessary to provide accurate and reliable metrics for cloud upload.
- **Hardware Mounting and Mechanical Challenges:**
 - Achieving consistent force measurement required precise alignment of the load cell with wooden plates and securing the structure to prevent uneven stress. Incorrect mounting initially caused reading fluctuations and force cancellation during off-center strikes.
 - Ensuring the kicking pad structure could withstand repeated strikes without damaging the load cell while maintaining sensitivity was a significant challenge, addressed through iterative design adjustments.

- **User and Prototype Constraints:**
 - Overly hard kicks risked damaging the 10 kg-rated load cell, limiting the range of forces that could be tested during development.
 - Maintaining continuous power to the Arduino Nano 33 IoT and Raspberry Pi was critical; accidental shutdowns temporarily stopped the system until reconnection occurred.

Despite these challenges, each issue was systematically addressed, resulting in a stable, functional prototype capable of measuring kick force, pressure distribution, and foot speed while providing real-time insights and feedback.

9. Future Improvements

The current Kickmeter prototype provides accurate measurement of kick force, pressure distribution, and foot speed, but several enhancements can further improve functionality, usability, and scalability:

- **Injury Warning System:** Implement real-time injury prevention alerts by monitoring unusual force patterns or joint stress during training sessions.
- **Posture and Kick Type Detection:** Integrate cameras with computer vision to capture posture and movement. Machine learning models can be trained to identify kick types (e.g., side kick, roundhouse kick, thrust kick) and detect incorrect form that may lead to injury.
- **Enhanced Data Visualization:** Add interactive graphs on the dashboard to track progress over time, visualize session statistics, and highlight trends in force, speed, and technique.
- **Improved Sensors:** Use higher-range and higher-precision sensors for the load cell and FSRs to achieve more accurate measurements and expand the detectable force range.
- **Expanded Social Features:** Upgrade the website to allow other practitioners to connect, share their progress, compare training sessions, and receive community feedback.
- **Multi-Martial Arts Application:** Extend the system beyond Taekwondo to other martial arts and combat sports, including disciplines where punches and hand strikes are primary, such as boxing or Muay Thai.

These improvements aim to make the Kickmeter not only a more comprehensive training tool but also a platform for community engagement, injury prevention, and advanced performance analytics across multiple martial arts.

10. Results

The Kickmeter system successfully captures and analyzes key metrics for Taekwondo training:

- **Kick Force Measurement:** Impact force is almost accurately measured in Newtons using the HX711 load cell.
 - **Speed Data:** Foot speed is reliably captured by the Arduino Nano 33 IoT accelerometer.
 - **Pressure Distribution:** FSR sensors record foot pressure across the kicking pad with consistent readings.
 - **Data Storage and Visualization:** Combined sensor data is stored in Firebase and visualized in real-time on the React.js dashboard.
 - **Performance Tracking:** The system provides quantitative feedback, allowing practitioners to monitor progress and improve technique over multiple sessions.
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11. Personal Journey

The development of the Kickmeter system was not only a technical challenge but also a significant learning experience:

- **Learning Curve:** Overcame challenges related to BLE communication, sensor calibration, and software integration.
 - **Persistence:** Iterative debugging and testing led to a stable and reliable system.
 - **Growth:** Gained hands-on experience in embedded systems design, asynchronous programming, and cloud integration.
 - **Satisfaction:** The project evolved from a conceptual idea to a functional training tool that combines Taekwondo practice with technology, offering measurable performance feedback.
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12. Conclusion and Reflection

The Kickmeter project successfully demonstrates the integration of hardware, software, and cloud analytics to enhance Taekwondo training. By combining force sensors, accelerometers, FSRs, and real-time data visualization, the system provides objective, quantitative feedback on kick performance that goes beyond traditional subjective observation.

Several challenges, including sensor calibration, BLE communication instability, and multi-source data synchronization, were systematically addressed through careful design, iterative testing, and continuous refinement. The resulting system allows practitioners to track their progress, analyze technique, and make data-driven improvements in real time.

Reflection:

- Developed practical expertise in embedded systems, sensor integration, and cloud-based data management.
- Gained experience in modular system design, multi-sensor fusion, and software-hardware co-development.
- Future enhancements could include machine learning for automatic kick classification, wireless FSR footpads, video-based posture and form analysis, and community features to support collaborative training.

Overall, the Kickmeter project highlights the importance of persistence, iterative development, and cross-disciplinary collaboration in creating functional, innovative, and user-focused training systems.

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